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AI-DRIVEN

CYBER SECURITY

YOUR AI CAN FOLLOW INSTRUCTIONS PERFECTLY, BUT WHAT HAPPENS WHEN A HACKER GIVES IT THE WRONG INSTRUCTIONS?

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PROMPT INJECTION

AN ATTACK WHERE MALICIOUS INSTRUCTIONS TRY TO MANIPULATE AN AI INTO IGNORING ITS INTENDED TASK OR SECURITY RULES.

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TYPES OF PROMPT INJECTION

- **DIRECT PROMPT INJECTION:** THE ATTACKER PUTS MALICIOUS INSTRUCTIONS DIRECTLY INTO THEIR MESSAGE, SUCH AS “IGNORE YOUR PREVIOUS INSTRUCTIONS.”
- **INDIRECT PROMPT INJECTION:** MALICIOUS INSTRUCTIONS ARE HIDDEN INSIDE EXTERNAL CONTENT THE AI READS, SUCH AS EMAILS, WEBSITES, PDFS, OR DOCUMENTS.

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SYSTEM PROMPT OVERRIDE

AN ATTEMPT TO MAKE THE AI IGNORE OR BYPASS ITS ORIGINAL SYSTEM INSTRUCTIONS. AN ATTACK MAY ATTEMPT THIS, BUT IT DOES NOT MEAN THE OVERRIDE WILL SUCCEED.

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TRUSTED VS. UNTRUSTED INPUT

SYSTEM INSTRUCTIONS DEFINE THE AI'S INTENDED BEHAVIOR, WHILE USER MESSAGES AND EXTERNAL CONTENT MUST BE TREATED CAREFULLY.

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PROMPT INJECTION VS. JAILBREAKING

PROMPT INJECTION MANIPULATES AN AI THROUGH ITS INPUTS OR CONTEXT.

JAILBREAKING ATTEMPTS TO BYPASS THE AI'S SAFETY RESTRICTIONS. THEY CAN OVERLAP.

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COMMON ATTACK GOALS

- REVEAL SENSITIVE INFORMATION, MISUSE CONNECTED TOOLS, GENERATE UNAUTHORIZED OUTPUTS, OR MANIPULATE AN AI AGENT INTO TAKING UNSAFE ACTIONS.
- WHY IT IS DANGEROUS: AI MAY ACCESS PRIVATE DOCUMENTS, DATABASES, APIS, OR TOOLS, SO SUCCESSFUL MANIPULATION CAN HAVE CONSEQUENCES BEYOND A SIMPLE TEXT RESPONSE.

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PREVENTION

- NEVER RELY ON PROMPTS ALONE. APPLY LEAST PRIVILEGE, VALIDATE INPUTS AND OUTPUTS, RESTRICT TOOL PERMISSIONS, REQUIRE HUMAN APPROVAL FOR SENSITIVE ACTIONS, AND TREAT RETRIEVED CONTENT AS UNTRUSTED.
- KEY PRINCIPLE: INSTRUCTIONS INSIDE USER CONTENT OR EXTERNAL DOCUMENTS MUST NOT AUTOMATICALLY BECOME TRUSTED COMMANDS.